

JPWM07 Light refraction — Quiz 1

S3 Physics · 3 multiple-choice items

Q1 · JPWM07.1 · Standard

The figure below shows a ray of light entering medium Y from medium X. Which of the following statements is/are correct?

- (1) Light travels faster in medium X than in medium Y.
- (2) ϕ is the angle of refraction.
- (3) The refractive index of medium X is larger than that of medium Y.

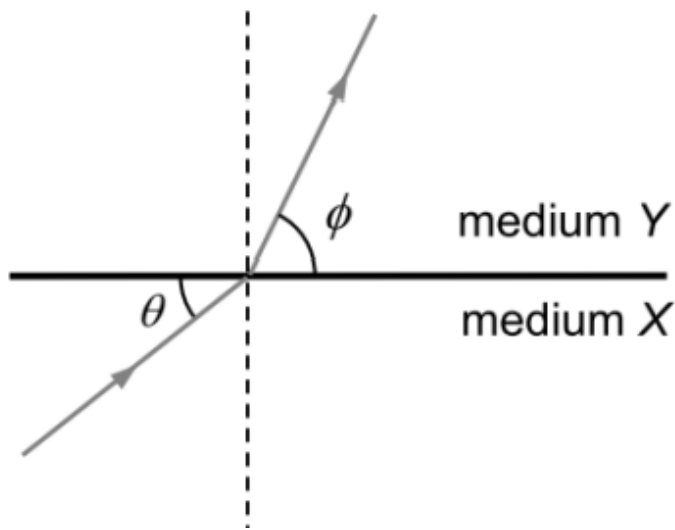


Fig · Refraction from medium X into medium Y

- **A.** (1) only
- **B.** (3) only
- **C.** (1) and (2) only
- **D.** (2) and (3) only

Q2 · JPWM07.2 · Standard

The given ray diagram shows the refraction of light at the surface of an unknown liquid. If the refractive index of the liquid is 1.52, what is the angle of incidence?

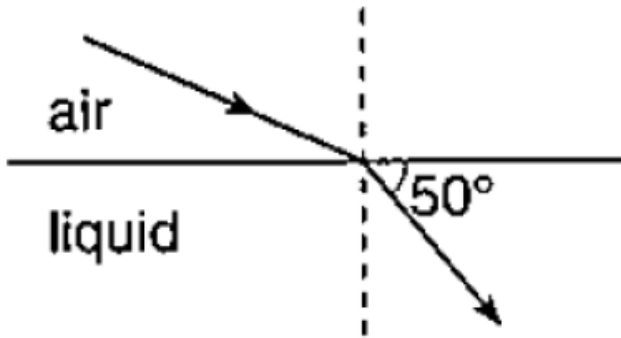


Fig · Refraction at an air–liquid surface

- **A.** 46.9°
- **B.** 52.6°
- **C.** 70°
- **D.** 77.7°

Q3 · JPWM07.3 · Foundation

The refractive index of diamond is 2.42. What is the speed of light in diamond? (Given that the speed of light in vacuum = $3 \times 10^8 \text{ ms}^{-1}$)

- **A.** $5.80 \times 10^7 \text{ ms}^{-1}$
- **B.** $1.24 \times 10^8 \text{ ms}^{-1}$
- **C.** $3.00 \times 10^8 \text{ ms}^{-1}$
- **D.** $7.26 \times 10^8 \text{ ms}^{-1}$

JPWM07 Light refraction — Quiz 1 · Answer key

Teacher page · 3 items

Q1 · JPWM07.1

Answer: A. (1) only

Guide: The light ray bends towards the normal when it travels from medium X to medium Y, so its speed decreases. Therefore (1) is correct. The angle of refraction is the angle between the refracted ray and the normal, so φ (measured from the surface) is not the angle of refraction and (2) is incorrect. By Snell's law, $n_X \sin \theta_X = n_Y \sin \theta_Y$. As $\theta_X > \theta_Y$ (angles from the normal), $n_X < n_Y$, so (3) is incorrect.

Q2 · JPWM07.2

Answer: D. 77.7°

Guide: The 50° mark is between the refracted ray and the surface, so the angle of refraction is $90^\circ - 50^\circ = 40^\circ$. Using Snell's law: $1 \times \sin i = 1.52 \times \sin 40^\circ$, so $\sin i \approx 0.977$ and $i \approx 77.7^\circ$.

Q3 · JPWM07.3

Answer: B. $1.24 \times 10^8 \text{ ms}^{-1}$

Guide: $n = c / v$, so $v = c / n = (3 \times 10^8 \text{ ms}^{-1}) / 2.42 \approx 1.24 \times 10^8 \text{ ms}^{-1}$.